

SS Credit: Rainwater Management

This credit applies to

- ▶ BD+C: New Construction (1-3 points)
- ▶ BD+C: Core and Shell (1-3 points)
- ▶ BD+C: Schools (1-3 points)
- ▶ BD+C: Retail (1-3 points)
- ▶ BD+C: Data Centers (1-3 points)
- ▶ BD+C: Warehouses and Distribution Centers (1-3 points)
- ▶ BD+C: Hospitality (1-3 points)
- ▶ BD+C: Healthcare (1-2 points)

INTENT

To reduce runoff volume and improve water quality by replicating the natural hydrology and water balance of the site, based on historical conditions and undeveloped ecosystems in the region to avoid contributing to flooding downstream in frontline communities.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Option 1. Percentile of Rainfall Events (1-3 points except Healthcare, 1-2 points Healthcare)

In a manner best replicating *natural site hydrology* processes, retain *on site* the runoff from the associated percentile of regional or local rainfall events. The percentile event volume must be retained (i.e. infiltrated, evapotranspired, or collected and reused) using *low-impact development (LID)* and *green infrastructure (GI) practices*. GI and LID strategies can be either structural or non-structural. Points are awarded according to Table 1.

Table 1. Points for percentile of rainfall retained

All Projects	Zero lot line Projects	Points	Points Healthcare
80 th Percentile	70 th Percentile	1	1
85 th Percentile	75 th Percentile	2	2
90 th Percentile	80 th Percentile	3	-

OR

Option 2. Natural land cover conditions (3 points except Healthcare, 2 points Healthcare)

Calculate the difference between the projected runoff volume under the proposed design conditions and the runoff volume under natural land cover conditions that existed prior to any disturbance. Retain (i.e. infiltrate, evapotranspire, or collect and reuse) on site the increase in runoff volume using LID and GI *practices*.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

Projects are now able to earn points for retaining runoff for the 80th, 85th, or 90th percentile of regional or local rainfall events, whereas the credit previously required the 95th or 98th percentile.

Thresholds for zero lot line projects have also been reduced. To ensure clarity, a more refined definition for zero lot line has been included and the term “manage” has been replaced with the more specific and technical “retain (i.e. infiltrate, evapotranspire, or collect and reuse)”. More guidance and resources have also been added to clarify acceptable LID strategies. New exemplary performance thresholds for all projects have been added, including exemplary performance under Option 2. Natural landcover conditions which rewards projects for fostering resilience by designing for future precipitation conditions.

Step-by-Step Guidance

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ The percentile rainfall event thresholds have been changed to 80th, 85th, and 90th percentile for all projects and 70th, 75th, and 80th percentile for zero lot line projects.
- ▶ Option 1, Step 2. Obtain Rainfall Data for Project Location: Use daily rainfall data and the methodology in the U.S. Environmental Protection Agency (EPA) Technical Guidance on Implementing the Stormwater Runoff Requirements for Federal Projects under Section 438 of the Energy Independence and Security Act to determine the percentile amount to be retained.
- ▶ Option 1, Step 2. Obtain Rainfall Data for Project Location: Once you’ve gathered the rainfall events data, any rainfall events less than or equal to 0.1 inches should be omitted.
- ▶ Option 1, Step 7. Conventional grey infrastructure devices and hard runoff conveyance systems, such as detention or retention ponds, pipes, culverts and vaults, can be compliant if used as a component of LID/GI systems to meet the requirements of this credit.

Further Explanation

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ Examples of acceptable LID techniques include the following:
 - planting rain gardens with native or adapted plant material (e.g. trees or shrubs);
 - installing a vegetated roof;
 - using permeable paving, consisting of porous above-ground materials (e.g., open pavers, engineered products), a base layer designed to drain water away from the building, and (often) a 6-inch-deep (150 millimeters) subbase; and
 - installing permanent infiltration or collection features (e.g., vegetated swale, rain garden, rainwater cistern) that can retain the runoff from the associated percentile of regional or local rainfall events.
- ▶ Treatment of Runoff:

Projects teams should consider prioritizing infiltration of runoff from impervious surfaces considered to be a significant source of pollutants in rainwater runoff. Such surfaces include those which are subject to: vehicular use; industrial activities; or storage of erodible or leachable materials, wastes, or chemicals and which receive direct rainfall or the run-on of rainfall.
- ▶ Green Infrastructure and Low-Impact Development Strategies:

Employ LID and GI strategies that mimic the natural infiltration-based, groundwater-driven hydrology of the site. If project teams are facing environmental or technical barriers to proceeding with LID techniques, consult the EPA’s Low-Impact Development website for detailed “Barrier Buster Fact Sheets” that address specific issues, such as how to combat clay soil constraints and designing LID practices on sloped sites.
- ▶ Project Type Variations, zero lot line projects:

Any setback area resulting from a planned below-grade space, such as an underground parking garage, does not void a project’s zero lot line status so long as the setback area is designed as pedestrian-oriented amenity space, such as outdoor seating.

Referenced Standards

- ▶ EPA’s LID website: <https://www.epa.gov/nps/lid>

- ▶ **Georgetown Climate Center's Green Infrastructure Toolkit:**
<http://www.georgetownclimate.org/adaptation/toolkits/green-infrastructure-toolkit/green-infrastructure-strategies-and-techniques.html>

Exemplary Performance

Option 1. Percentile of rainfall events

All Projects	Zero lot line Projects
98 th Percentile	95 th Percentile

Option 2. Natural land cover conditions

Retain on site an additional 20% in runoff volume to address future precipitation rates (1.2 * [difference in annual runoff volume from the natural landcover to the post-developed condition]) using LID and GI practices.

Definitions

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ **retain (rainwater) on site** to capture and reserve a specified volume of rainfall to mimic natural hydrologic function. Retention is a function of rainwater management that includes strategies involving evapotranspiration, infiltration, and capture and reuse.

Connection to Ongoing Performance:

- ▶ LEED O+M WE credit Water Performance: LID and GI measures that harvest and reuse rainwater may help reduce potable water demand. Thus, these measures help ensure that projects will use less water throughout the building life cycle, which may help improve a project's water performance score. Tracking water consumption on a regular basis through metering supports effective water management and provides performance data to help verify that building systems are operating as designed.
- ▶ LEED O+M SS credit Heat Island Reduction: Vegetated roofs installed for achievement of this credit will also qualify for the related performance-based credit.